

- 1 (a) A progressive wave is described by the equation $y = 0.5 \sin \left[\pi \left(\frac{t}{5} + 5x \right) \right]$, where x and y are in meter and t is in second. Determine the
- (i) amplitude
 - (ii) angular velocity
 - (iii) wave number.
 - (iv) frequency
 - (v) wavelength.
 - (vi) wave propagation velocity.
- [9 marks]
- (b) A rope of mass 220g and length 1.6 m has one of its end fixed to a wall and the other end is forced to vibrate transversely with a frequency of 30 Hz. If the rope is stretched to 120 N, calculate the
- (i) mass per unit length of the rope
 - (ii) velocity of the wave
 - (iii) wavelength.
- [7 marks]
- 2 (a) 0.80 A current flows in a uniform wire of length 3.5 m and radius of 0.5 mm when a potential difference of 10 V is applied to the wire. Calculate the
- (i) resistance of the wire
 - (ii) resistivity of the wire
- [4 marks]
- (b) A silver wire carries a current of 5.0 A. Determine the number of electrons pass through the wire in 1 hour.
($e = 1.60 \times 10^{-19} \text{ C}$)
- [3 marks]
- (c) A potential difference of 25 V is applied across the two ends of a wire of resistance 3.5Ω , diameter 1.5 mm and resistivity of $2.5 \times 10^{-5} \Omega \text{ m}$. Determine the length of the wire.
- [2 marks]

(d)

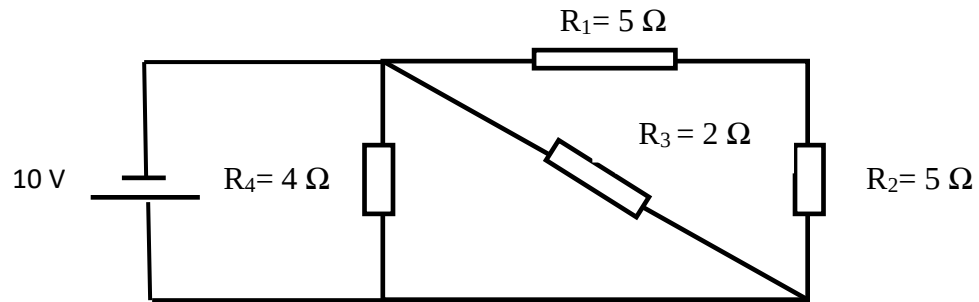
**FIGURE 2**

FIGURE 2 shows a circuit consists of resistors connected to a power supply. Calculate the equivalent resistance of the circuit.

[3 marks]

3 (a)

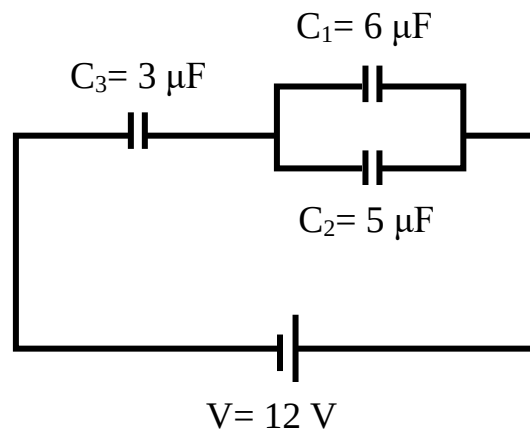
**FIGURE 3**

FIGURE 3 shows an arrangement of three capacitors connected to a 12 V battery. Calculate the

- (i) effective capacitance
- (ii) potential difference across capacitor C_1 if its charge is $50\ \mu\text{C}$

[5 marks]

(b) A 12 V battery is connected to a $6\ \mu\text{F}$ capacitor and a $35\ \Omega$ resistor. Calculate the

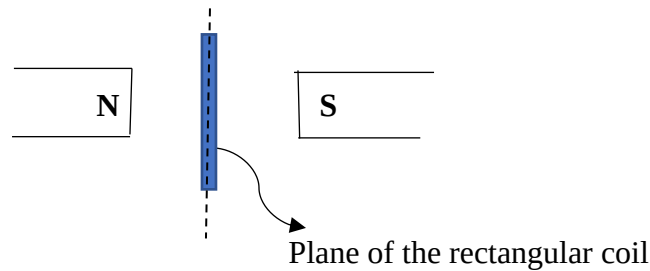
- (i) maximum charge stored
- (ii) time constant
- (iii) charge at time $t = 0.42\ \text{ms}$.

[5 marks]

4. (a) (i) A current of 3.5 A flows in a long straight conductor in free space. Determine the magnitude of the magnetic field at a point which is at a perpendicular distance of 2.0 cm from the conductor.
- (ii) A solenoid of 30 turns is carrying a current of 20 mA. The magnetic field at the centre is 1.5×10^{-6} T. Calculate the length of the solenoid.
(Given: $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$)

[4 marks]

(b)

**FIGURE 4**

The **FIGURE 4** shows the plan view of a rectangular coil in magnetic field. The area of the rectangular coil is 6 cm^2 and the magnetic field strength is 2.5 T.

- (i) Calculate the magnetic flux through the coil when the position of the coil is as shown in **FIGURE 4**.
- (ii) Calculate the change in the magnetic flux through the coil when the angle of the plane of the coil and magnetic field is at 60° .

[5 marks]

- (c) A circular loop of radius 3 cm is placed in a uniform magnetic field of magnitude 0.650 T. The normal to the loop makes an angle of 30.0° with respect to the direction of B . If the field increases to 0.8 T in 0.5 s, calculate

- (i) the induced emf during this time interval.
- (ii) induced current if the loop have resistance of 8Ω .

[5 marks]

5. (a) A convex mirror of focal length 15.0 cm produces a virtual image 8.0 cm from the mirror.
- (i) Sketch a labeled ray diagram to show the formation of the image.
- (ii) Calculate the object distance.

[4 marks]

- (b) A concave mirror of focal length 10 cm produces a virtual image with triple the size of the object. Calculate the
- (i) object distance
 - (ii) position of the image distance.

[5 marks]

- (c) An object of height 5 cm is placed 25.0 cm from a converging lens that has a focal length of 8.0 cm. Determine the
- (i) position of the image,
 - (ii) height of the image.

[5 marks]

6. (a) The separation of a double-slit is 0.25 mm. A beam of monochromatic light of wavelength 500 nm is incident on this double-slit. If the screen is located 0.5 m away from the slit, calculate the distance from the central bright fringe to the
- (i) third bright fringe
 - (ii) second dark fringe

[6 marks]

- (b) (i) A Young's double slit experiment is performed with 600 nm light source and the distance between slits and screen is 1.5 m. The tenth minimum is observed at 8.25 mm from the central maximum. Determine the spacing between the slits.
- (ii) Calculate the fringe separation
 - (iii) Explain in a double-slit interference pattern if the experiment were submerged in water.

[8 marks]